

Review3

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Review Questions

Part 3

These are review questions for continuous variable (Chapters 5, 6 in Ross' book).

(1) For some constant c , the random variable X has the probability density function

$$f(x) = \begin{cases} cx^4 & 0 < x < 2; \\ 0 & \text{otherwise.} \end{cases}$$

Find (a) $\mathbb{E}(X)$ and (b) $\text{Var}(X)$.

solving for c , $\int_0^2 cx^4 dx = 1$

$$c \left[\frac{x^5}{5} \right]_0^2 = 1$$

$$c \left[\frac{(2)^5}{5} - 0 \right] = 1$$

$$c = \frac{5}{32}$$

$$(a) \mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_0^2 x \left(\frac{5}{32} x^4 \right) dx$$

$$= \frac{5}{32} \int_0^2 x^5 dx$$

$$= \frac{5}{32} \left[\frac{x^6}{6} \right]_0^2$$

$$= \frac{5}{32} \left[\frac{(2)^6}{6} - 0 \right]$$

$$= \frac{5}{3}$$

$$(b) \mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^2 x^2 \left(\frac{5}{32} x^4 \right) dx = \frac{5}{32} \left[\frac{x^7}{7} \right]_0^2$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= \frac{20}{7} - \left(\frac{5}{3} \right)^2 = \frac{5}{63} \end{aligned}$$

(2) The random variable X has the probability density function

$$f(x) = \begin{cases} ax + bx^2 & 0 < x < 1; \\ 0 & \text{otherwise.} \end{cases}$$

If $\mathbb{E}(X) = 0.6$, find (a) $P\{X < \frac{1}{2}\}$, and (b) $\text{Var}(X)$.

(a) First, solve for a and b , using $\mathbb{E}(X) = 0.6$ and $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$\int_0^1 (ax + bx^2) dx = 1$$

$$\left[\frac{ax^2}{2} + \frac{bx^3}{3} \right]_0^1 = 1$$

$$\frac{a}{2} + \frac{b}{3} = 1$$

$$3a + 2b = 6 \quad \text{--- (1)}$$

Solving (1) and (2), $a = 3.6$, $b = -2.4$.

$$\int_0^1 x(ax + bx^2) dx = 0.6$$

$$\int_0^1 (ax^2 + bx^3) dx = 0.6$$

$$\left[\frac{ax^3}{3} + \frac{bx^4}{4} \right]_0^1 = 0.6$$

$$\frac{a}{3} + \frac{b}{4} = 0.6$$

$$4a + 3b = 7.2 \quad \text{--- (2)}$$

$$\begin{aligned} P(X < \frac{1}{2}) &= \int_0^{0.5} (3.6x - 2.4x^2) dx \\ &= \left[\frac{3.6}{2}x^2 - \frac{2.4}{3}x^3 \right]_0^{0.5} \\ &= 0.35, \end{aligned}$$

$$\begin{aligned} \text{(b) } \mathbb{E}(X^2) &= \int_0^1 x^2 (3.6x - 2.4x^2) dx \\ &= \left[\frac{3.6x^4}{4} - \frac{2.4x^5}{5} \right]_0^1 \\ &= \frac{3.6}{4}(1)^4 - \frac{2.4}{5}(1)^5 \\ &= 0.42 \end{aligned}$$

$$\begin{aligned} \therefore \text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= 0.42 - 0.6^2 \\ &= 0.06, \end{aligned}$$

- (3) The annual rainfall in Cleveland, Ohio is approximately a normal random variable with mean 40.2 inches and standard deviation 8.4 inches. What is the probability that
- (a) next year's rainfall will exceed 44 inches?
- (b) the yearly rainfalls in exactly 3 of the next 7 years will exceed 44 inches?
- Assume that if A_i is the event that the rainfall exceeds 44 inches in year i (from now), then the events $A_i, i \geq 1$, are independent.

(a) Let X be the r.v. denoting the annual rainfall in inches.

$$X \sim N(40.2, 8.4^2) \text{ approx.}$$

$$P(X > 44) = P\left(\frac{X - 40.2}{8.4} > \frac{44 - 40.2}{8.4}\right)$$

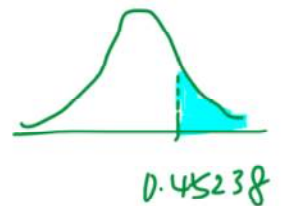
$$= P(Z > 0.45238)$$

$$= 1 - \Phi(0.45238)$$

$$= 1 - 0.6736$$

$$\approx 0.3264,$$

Note = exact answer computed is 0.3255



(b) Let Y be the r.v. denoting the yearly rainfall exceeds 44 inches.

$$Y \sim \text{Bin}(7, 0.3264) \quad \leftarrow \text{Due to independence Assumption.}$$

$$P(Y=3) = \binom{7}{3} (0.3264)^3 (1-0.3264)^{7-3}$$

$$\approx 0.2506,$$

(4) A roulette wheel has 38 slots, numbered 0, 00, and 1 through 36. If you bet 1 on a specified number then you either win 35 if the roulette ball lands on that number or lose 1 if it does not. If you continually make such bets, approximate the probability that

(a) you are winning after 34 bets;

(b) you are winning after 1000 bets;

(c) you are winning after 100,000 bets.

Assume that each roll of the roulette ball is equally likely to land on any of the 38 numbers.

(a) Let p be the profit/loss gathered, i is the # of times ball land on '1'.

$$p = 35i - 1(34 - i)$$

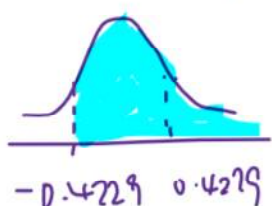
$$p = 36i - 34$$

$$36i - 34 > 0 \Rightarrow i > \frac{34}{36}$$

$$\Rightarrow i > 1 \quad (\text{since } i \text{ is an integer})$$

Let x be the r.v. denoting the # of '1's. $X \sim \text{Bin}(34, \frac{1}{38})$

$$P(X \geq 1) = P(X \geq 0.5) \quad [\text{Binomial} \rightarrow \text{Normal approx.}]$$



$$= P\left(\frac{X - 34(\frac{1}{38})}{\sqrt{34(\frac{1}{38})(\frac{37}{38})}} \geq \frac{0.5 - 34(\frac{1}{38})}{\sqrt{34(\frac{1}{38})(\frac{37}{38})}}\right)$$

$$= P(Z \geq -0.4229) = \Phi(0.4229)$$

$$(b) p = 35i - (1000 - i) = 36i - 1000 \geq 0 \Rightarrow i \geq \frac{1000}{36} \approx 27.78 \Rightarrow i \geq 28 \text{ games.}$$

$$X \sim \text{Bin}(1000, \frac{1}{38}), \quad P(X \geq 28) = P(X \geq 27.5)$$

$$= P\left(Z \geq \frac{27.5 - \frac{1000}{38}}{\sqrt{1000(\frac{1}{38})(\frac{37}{38})}}\right)$$

$$(c) p = 35i - (100000 - i) = 36i - 100000 \geq 0 \Rightarrow i \geq 2778 \text{ games.}$$

$$= P(Z \geq 0.2339)$$

$$P(X \geq 2778) = P(X \geq 2777.5)$$

$$= 1 - \Phi(0.2339)$$

$$= P(Z \geq 0.2883) \approx 0.0021$$

$$\approx 1 - 0.5910 = 0.4090$$

- (5) Let X be the number of times that a fair coin that is flipped 40 times lands on heads. Find the probability that $X = 20$, by using the normal approximation.

$$X \sim \text{Bin}(40, \frac{1}{2})$$

$$P(X=20) = P(19.5 \leq X \leq 20.5) \quad \text{Normal Approx.}$$
$$= P(X \leq 20.5) - P(X \leq 19.5)$$

$$= P\left(Z \leq \frac{20.5 - 40(\frac{1}{2})}{\sqrt{40(\frac{1}{2})(\frac{1}{2})}}\right) - P\left(Z \leq \frac{19.5 - 40(\frac{1}{2})}{\sqrt{40(\frac{1}{2})(\frac{1}{2})}}\right)$$

$$= P(Z \leq 0.15811) - P(Z \leq -0.15811)$$

$$= \Phi(0.15811) - (1 - \Phi(0.15811))$$

$$= 2\{\Phi(0.15811)\} - 1$$

$$= 2(0.5636) - 1$$

$$\approx 0.1272$$

- (6) To determine the effectiveness of a certain diet in reducing the amount of cholesterol in the blood-stream, 100 people are put on the diet. After they have been on the diet for a sufficient length of time, their cholesterol count will be taken. The nutritionist running this experiment has decided to endorse the diet if at least 65 percent of the people have a lower cholesterol count after going on the diet. What is the probability that the nutritionist endorses the new diet if, in fact, it has no effect on the cholesterol level?

* Assumption: There is a fair chance that the new diet either lowers or does not lower the cholesterol level of the participants.

$X \sim \text{Bin}(100, \frac{1}{2})$, where X is the # of people that have a lower cholesterol count after the new diet.

$$\begin{aligned} P(X \geq 65) &= P(X \geq 64.5) \quad (\text{Bin} \sim \text{Normal approx.}) \\ &= P\left(Z \geq \frac{64.5 - 100(\frac{1}{2})}{\sqrt{100(\frac{1}{2})(\frac{1}{2})}}\right) \\ &= P(Z \geq 2.9) \\ &= 1 - \Phi(2.9) \\ &\approx 1 - 0.9981 \\ &= 0.0019, \end{aligned}$$

- (7) Suppose that the number of miles that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 miles. If a person desires to take a 5000-mile trip, what is the probability that he or she will be able to complete the trip without having to replace the car battery? What can be said when the distribution is not exponential?

$$E(X) = \frac{1}{\lambda}$$

$$10,000 = \frac{1}{\lambda}$$

$$\lambda = \frac{1}{10000}$$

$$F(x) = 1 - e^{-\lambda x} \rightarrow P(X \leq x)$$

$$= 1 - e^{-\frac{1}{10000}x}$$

$$P(X > 5000) = 1 - F(5000)$$

$$= 1 - \left(1 - e^{-\frac{5000}{10000}}\right)$$

$$= e^{-\frac{1}{2}}$$

$$\approx 0.6065,$$

depends on when it was last used.

If the lifetime distribution F is not exponential, then the relevant probability is

$$P\{\text{lifetime} > t + 5 \mid \text{lifetime} > t\} = \frac{1 - F(t + 5)}{1 - F(t)},$$

where t is the number of miles that the battery had been in use prior to the start of the trip.

Conclusion: if the distribution is not exponential, additional information is needed (namely, the value of t) before the desired probability can be calculated.

Exponential is memoryless : $P(L > t + 5 \mid L > t) = \frac{P(L > t + 5)}{P(L > t)}$

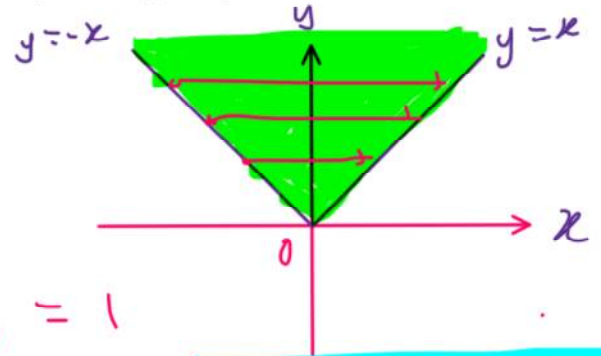
$$= e^{-5\lambda} = \frac{1 - (1 - e^{-\lambda(t+5)})}{1 - e^{-\lambda t}}$$

(8) The joint density function of X and Y is given by

$$f(x, y) = C(y - x)e^{-y}, \quad -y < x < y, 0 < y < \infty$$

and is equal to 0 outside the region.

- (a) Find C .
- (b) Find the density function of X .
- (c) Find the density function of Y .
- (d) Find $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.



$$(a) \int_0^{\infty} \int_{-y}^y C(y-x)e^{-y} dx dy = 1$$

$$C \int_0^{\infty} e^{-y} \left[yx - \frac{x^2}{2} \right]_{-y}^y dy = 1$$

$$C \int_0^{\infty} e^{-y} \left[(y^2 - \frac{1}{2}y^2) - (-y^2 - \frac{1}{2}y^2) \right] dy = 1$$

$$2C \int_0^{\infty} y^2 e^{-y} dy = 1$$

$$C = \frac{1}{4}$$

Useful formula;

$$\int_0^{\infty} e^{-y} y^n dy = n!$$

For $n \in \mathbb{Z}^+$

Do IBP 2 times...

(b) If $x \geq 0$: $f_x(x) = \int_x^{\infty} \frac{1}{4}(y-x)e^{-y} dy$, let $u = y-x$, $\frac{du}{dy} = 1$

$$= \frac{1}{4} \int_0^{\infty} ue^{-(u+x)} du$$

$$= \frac{1}{4} e^{-x} \int_0^{\infty} ue^{-u} du = \frac{1}{4} e^{-x} \text{ if } x \geq 0$$

If $x < 0$: $f_x(x) = \int_{-x}^{\infty} \frac{1}{4}(y-x)e^{-y} dy$

$$= \frac{1}{4} \left[[-(y-x)e^{-y}]_{-x}^{\infty} - \int_{-x}^{\infty} -e^{-y} dy \right]$$

$$= \frac{1}{4} [-2xe^x - [e^{-y}]_{-x}^{\infty}]$$

$$= \frac{1}{4} [-2xe^x + e^x] \text{ if } x < 0$$

(8) The joint density function of X and Y is given by

$$f(x, y) = C(y - x)e^{-y}, \quad -y < x < y, 0 < y < \infty$$

and is equal to 0 outside the region.

- (a) Find C .
- (b) Find the density function of X .
- (c) Find the density function of Y .
- (d) Find $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

$$\begin{aligned} \text{(c) } f_Y(y) &= \int_{-y}^y \frac{1}{4}(y-x)e^{-y} dx \\ &= \frac{1}{4}e^{-y} \left[xy - \frac{x^2}{2} \right]_{-y}^y \\ &= \frac{1}{4}e^{-y} \left[(y^2 - \frac{y^2}{2}) - (-y^2 - \frac{y^2}{2}) \right] \\ &= \frac{1}{2}y^2e^{-y} \text{ if } y > 0, // \end{aligned}$$

(d)

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_{-\infty}^0 x \left[\frac{1}{4}(-2xe^x + e^x) \right] dx + \int_0^{\infty} x \left[\frac{1}{4}e^{-x} \right] dx$$

$$= \int_{-\infty}^0 -\frac{1}{2}x^2e^x dx + \int_{-\infty}^0 \frac{1}{4}xe^x dx + \int_0^{\infty} \frac{1}{4}xe^{-x} dx$$

let $u = -x$ ↓

↓ let $u = -x$

$$= \frac{1}{2} \int_{\infty}^0 -u^2e^{-u}(-du) + \frac{1}{4} \int_{\infty}^0 -ue^{-u}(-du) + \frac{1}{4} \int_0^{\infty} xe^{-x} dx$$

$$= -\frac{1}{2} \int_0^{\infty} u^2e^{-u} du - \frac{1}{4} \int_0^{\infty} ue^{-u} du + \frac{1}{4} \int_0^{\infty} xe^{-x} dx$$

$$= -\frac{1}{2}(2!) - \frac{1}{4} + \frac{1}{4} \quad \leftarrow \text{Useful Formula.}$$

$$= -1, //$$

$$\begin{aligned} \mathbb{E}(Y) &= \int_0^{\infty} y \left(\frac{1}{2}y^2e^{-y} \right) dy = \frac{1}{2} \int_0^{\infty} y^3e^{-y} dy = \frac{1}{2}(3!) \\ &= 3, // \end{aligned}$$

(9) . The joint density function of X and Y is given by

$$f(x, y) = xy, \quad 0 < x < 1, 0 < y < 2$$

and is equal to 0 outside the region.

- Are X and Y independent?
- Find the density function of X .
- Find the density function of Y .
- Find the joint distribution function.
- Find $E(Y)$.
- Find $P\{X + Y > 1\}$.

(a) Yes. $f(x, y)$ factors as x and y

and $0 < x < 1, 0 < y < 2$.

$$(b) f_X(x) = \int_0^2 xy \, dy = \left[\frac{xy^2}{2} \right]_0^2$$

$$= 2x, \quad 0 < x < 1$$

$$(c) f_Y(y) = \int_0^1 xy \, dx = \left[\frac{yx^2}{2} \right]_0^1 = \frac{y}{2}, \quad 0 < y < 2.$$

$$(d) F_{XY}(x \leq x, y \leq y) \quad \text{"x" "y"}$$

$$= \int_{-\infty}^y \int_{-\infty}^x f(u, v) \, du \, dv$$

$$= \int_{-\infty}^y \int_{-\infty}^x uv \, du \, dv$$

$$= \int_0^y u \, du \int_0^x v \, dv$$

$$= \left[\frac{u^2}{2} \right]_0^y \left[\frac{v^2}{2} \right]_0^x$$

$$= \frac{y^2}{2} \times \frac{x^2}{2}$$

$$= \frac{1}{4} x^2 y^2, \quad 0 < x < 1, 0 < y < 2.$$

$$= \min\{1, x^2\} \min\{1, \frac{y^2}{4}\} \quad \text{for } x > 0, y > 0.$$

$$(e) E(Y) = \int_0^2 y \left(\frac{y}{2}\right) dy = \frac{1}{2} \left[\frac{y^3}{3} \right]_0^2 = \frac{4}{3} //$$

$$(f) P\{X+Y > 1\}$$

$$= P\{Y > 1-X\}$$

$$= \int_0^1 \int_{1-x}^2 xy \, dy \, dx$$

$$= \int_0^1 \left[\frac{xy^2}{2} \right]_{1-x}^2 dx$$

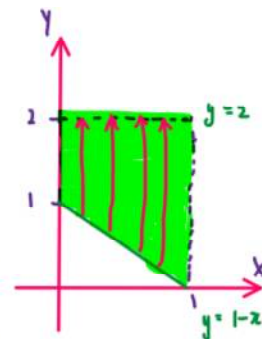
$$= \frac{1}{2} \int_0^1 4x - x(1-x)^2 dx$$

$$= \frac{1}{2} \int_0^1 4x - x + 2x^2 - x^3 dx$$

$$= \frac{1}{2} \left[\frac{3}{2} x^2 + \frac{2}{3} x^3 - \frac{x^4}{4} \right]$$

$$= \frac{1}{2} \left[\frac{3}{2} + \frac{2}{3} - \frac{1}{4} \right]$$

$$= \frac{23}{24} //$$



- (10) Let N be a geometric random variable with parameter p . Suppose that the conditional distribution of X given that $N = n$ is the gamma distribution with parameters n and λ . Find the conditional probability mass function of N given that $X = x$.

$$\begin{aligned}
 P_{N|X=x}(N|X=x) &= \frac{P_{NX}(n, x)}{P_X(x)} \\
 &= \frac{P_{X|N}(x|n) P_N(n)}{P_X(x)}
 \end{aligned}$$

$$N \sim \text{Geom}(p)$$

$$P_N(n) = (1-p)^{n-1} p$$

$$P_{X|N}(x|n) = \text{Gamma}(n, \lambda)$$

$$= \frac{\lambda^n}{\Gamma(n)} x^{n-1} e^{-\lambda x}$$

$$\text{Since } n \in \mathbb{Z}^+, \Gamma(n) = (n-1)!$$

$$\therefore P_{N|X=x}(N|X=x) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x} (1-p)^{n-1} p}{P_X(x)}$$

$$= \frac{p\lambda}{P_X(x)} \cdot \frac{\lambda^{n-1}}{(n-1)!} (1-p)^{n-1} e^{-\lambda x}$$

$$P_X(x) = p\lambda e^{-p\lambda x}$$

$$X \sim \text{Exp}(p\lambda).$$

$$= \frac{p\lambda e^{-p\lambda x}}{p\lambda e^{-p\lambda x}} \cdot \frac{[\lambda(1-p)]^{n-1} e^{-\lambda(1-p)x}}{(n-1)!}$$

$$= P\{Y = n-1\}, \quad Y \sim \text{Poisson}(\lambda(1-p)), n \geq 1.$$